IE643 Lecture 7

Announcement:

Quiz-1. Date: 1st September 2026 (Tuesday) Venue: In class. Syllabus: Contents till Perceptron mistake bound analysis. Open scribes/notes. More instructions to be posted in moodle.

Recap:

$$x \in \mathbb{R}^d$$

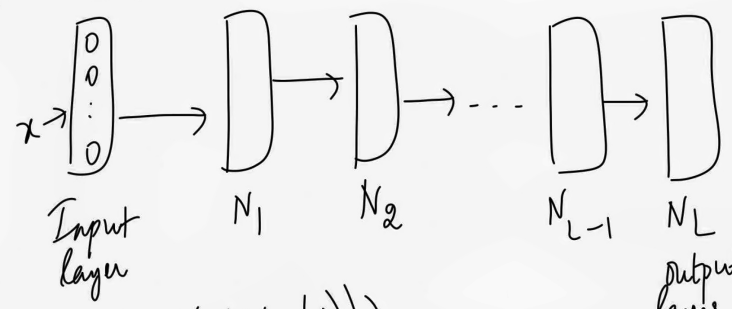


Figure 1: Input layer (column of nodes, $x \rightarrow$) feeding through layers $N_1, N_2, \ldots, N_{L-1}, N_L$ (output layer), each connected to the next by an arrow.

$$\hat{y} = \vec{a}^L = \phi(W^L \phi(W^3 \phi(W^2 \phi(W^1 x))))$$

$e(y, \hat{y}) \rightarrow$ loss fn to compare true and ...

Supervised Learning

Supervised learning: $D = \{(x^i, y^i)\}_{i=1}^N$

$$X \ni x^i \in \mathbb{R}^d$$
$$Y \ni y^i \in \{+1, -1\}$$

$h : X \rightarrow Y$, MLP weights

train a multilayered perceptron to find h_{MLP}

$$\min E = \sum_{i=1}^{N} e(y^i, \hat{y}^i)$$

minimization is over weights of MLP.

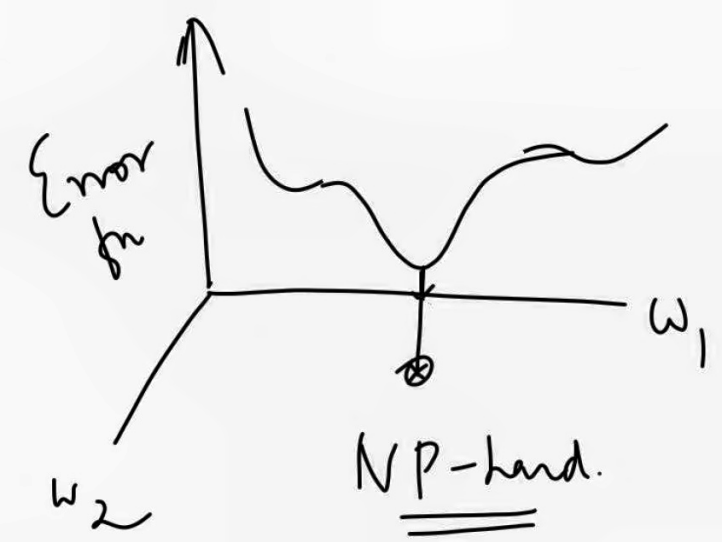


Figure 2: A wavy error-surface plot over axes w_1, w_2 with a marked local minimum.

local minimum, **NP-hard.**

Local Minimum

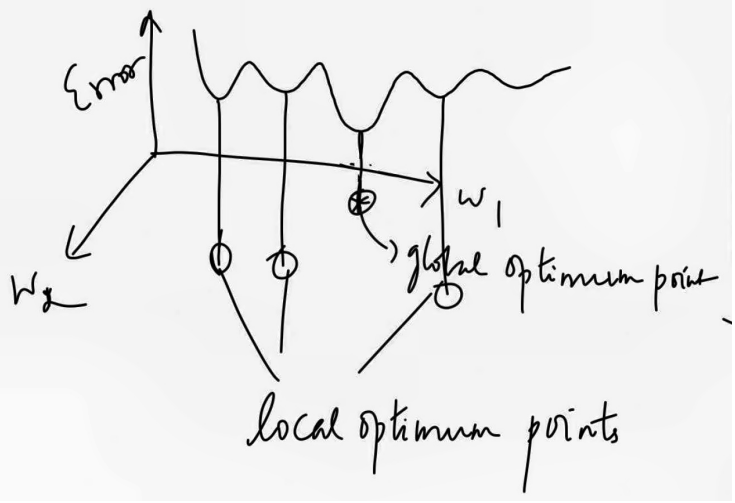


Figure 3: A jagged error curve over w_1, w_2 with several marked local optimum points and one marked “global optimum point.”

local optimum points, global optimum point

Iterative procedure to find a local minimum point

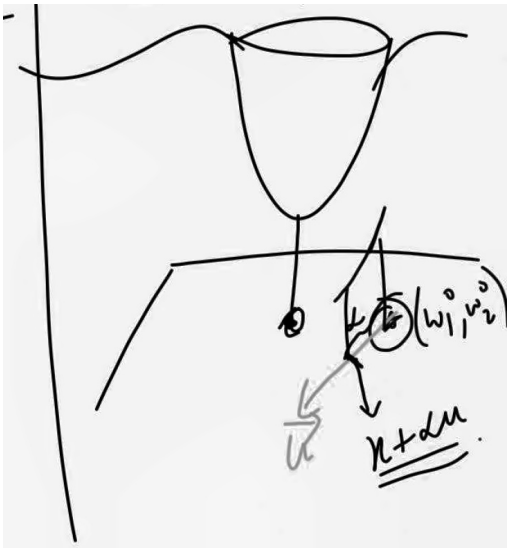


Figure 4: A bowl-shaped surface with a ball rolling down toward the minimum, labeled with weight coordinates (w_1^0, w_2^0) and an update arrow toward $x + \alpha u$.

Directional derivative.

For a fn. $f : \mathbb{R}^d \to \mathbb{R}$, the directional derivative along some non-zero direction $u \in \mathbb{R}^d$ at a point x is defined as

$$f'(x; \vec{u}) = \lim_{\alpha \downarrow 0} \frac{f(x + \alpha u) - f(x)}{\alpha}$$

$\nabla f(x)$ at a point x:

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix}$$

$$f'(x; u) = \langle \nabla f(x), u \rangle$$

$$u = -\nabla f(x)$$

If $\nabla f(x) \neq$ zero vector: $f'(x; u) = \langle \nabla f(x), u \rangle = -\|\nabla f(x)\|^2$

Descent Direction

$$f(x; -\nabla f(x)) < 0$$

Descent direction:

A non-zero direction $u \in \mathbb{R}^d$ is called a descent direction of f at x if:

$$f'(x; u) < 0$$

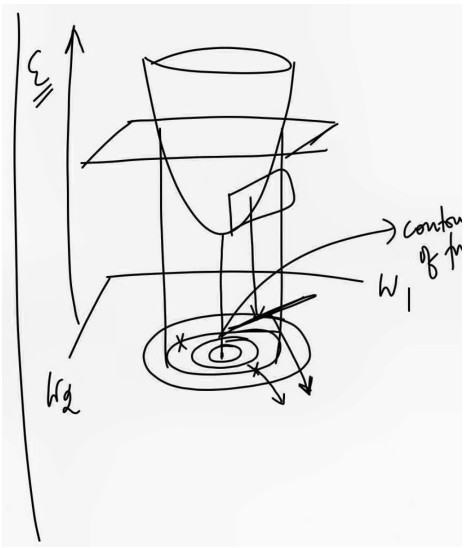


Figure 5: A 3D bowl-shaped (paraboloid) surface over axes w_1, w_2 with a contour-plot base showing concentric circles and gradient-direction arrows pointing toward the center (minimum).

Negative gradient direction $-\nabla f(x)$

Step-length/learning rate.

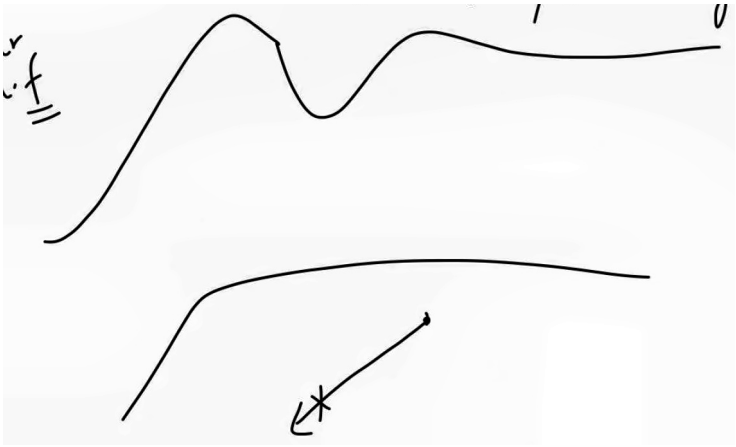


Figure 6: A wavy curve illustrating a step taken from one point to another along the curve, with an arrow marking the direction of the step.

$$f'(x; u) = \lim_{\alpha \downarrow 0} \frac{f(x + \alpha u) - f(x)}{\alpha}$$

Property: If u (a non-zero vector) is a descent direction of fn $f : \mathbb{R}^d \to \mathbb{R}$ at x, then:

there exists some $\epsilon \geq 0$ such that for every $\alpha \in (0, \epsilon]$,

$$f(x + \alpha u) < f(x)$$

$$\lim_{\alpha \downarrow 0} \frac{f(x + \alpha u) - f(x)}{\alpha} < 0 \iff f'(x; u) < 0$$

Gradient Descent

Gradient descent to solve

$$\min_{x \in \mathbb{R}^d} f(x), \qquad f : \mathbb{R}^d \to \mathbb{R}$$

1. local minimum
2. saddle points ??

$\to$ Start at $x^0 \in \mathbb{R}^d \to$ for $k = 0, 1, 2, ...$ $\hookrightarrow$ find a descent direction u^k (−ve gradient) $\hookrightarrow$ find step length α_k $\hookrightarrow x^{k+1} \leftarrow x^k + \alpha_k u^k$

$$\min_W E = \sum_{i=1}^{N} e(y^i, \hat{y}^i) \qquad \nabla E = \sum_{i=1}^{N} \nabla e(y^i, \hat{y}^i)$$

N: very large $\to$ over... (overload)

full gradient vs. sample-based gradient

- w^0
- for $k = 0, 1, 2, ...$
  - find descent dirn using $-\nabla E(w^k)$
  - find a suitable step length $\alpha_k \to$ step-length finding procedures ($10^{-5}, 10^{-4}, ...$; α_k fixed to be small value close to ...; 10^{-8})
  - $w^{k+1} \leftarrow w^k + \alpha_k(-\nabla E(w^k))$
  - zero-gradient check to stop this process

Stochastic gradient descent.

$$D = \{(x^i, y^i)\}_{i=1}^N$$

$\to w^0 \to$ for $k = 0, 1, 2, ...$ $\hookrightarrow$ Compute a stochastic gradient randomly choose $j \in \{1, 2, 3, ... N\}$ Compute $\nabla e(y^j, \hat{y}^j)$

$$w^{k+1} \leftarrow w^k + \eta_k(-\nabla e(y^j, \hat{y}^j))$$

Stochastic approx.'[51] $\to$ convergence to the local opt/saddle pt $\to$ might not lead to descent

Mini-batch SGD

$$D = \{(x^i, y^i)\}_{i=1}^N$$

$\rightarrow w^0 \rightarrow$ for $k = 0, 1, 2, ...$ $\hookrightarrow$ Sample mini-batch of samples $B_k \subseteq \{1, 2, ... N\}$ for every $j \in B_k$: compute $\nabla e(y^j, \hat{y}^j)$ $\hookrightarrow w^{k+1} \leftarrow w^k + \eta_k \left(-\sum_{j \in B_k} \nabla e(y^j, \hat{y}^j)\right)$

convergence in asymptotic sense

Adam. SGD with momentum

$$m_k = \alpha \cdot \text{grad}_k + (1 - \alpha) m_{k-1}$$

generalizn perf

$$u^k = \alpha(\text{mini-batch gradient}) + (1 - \alpha) \cdot \text{past gradients.}$$

Adam: AMSgrad

$$w^{k+1} \leftarrow w^k + \eta_k \left(\frac{-\text{running avg of gradients}}{\sqrt{\text{running avg of sq. comp of gradients}} + \epsilon} \right)$$

$$m_k = \beta_1 \cdot \text{grad}_k + (1 - \beta_1) m_{k-1}$$
$$v_k = \beta_2 \cdot (\text{grad}_k^2) + (1 - \beta) v_{k-1}$$

$$w^{k+1} \leftarrow w^k + \eta_k \left(-\frac{m_k}{\sqrt{v_k} + \epsilon} \right)$$

Sample-based gradient

$$\nabla e(y^j, \hat{y}^j)$$

Computation of sample-based gradient

$$\hat{y}^j = \phi(W^L \cdots \phi(W^2 \phi(W^1 x^j)) \cdots)$$

$$\nabla_1 e(y^j, \hat{y}^j)$$

$$w^1, w^2, w^3, \ldots, w^L$$

auto-differentiation

Forward Chain Rule

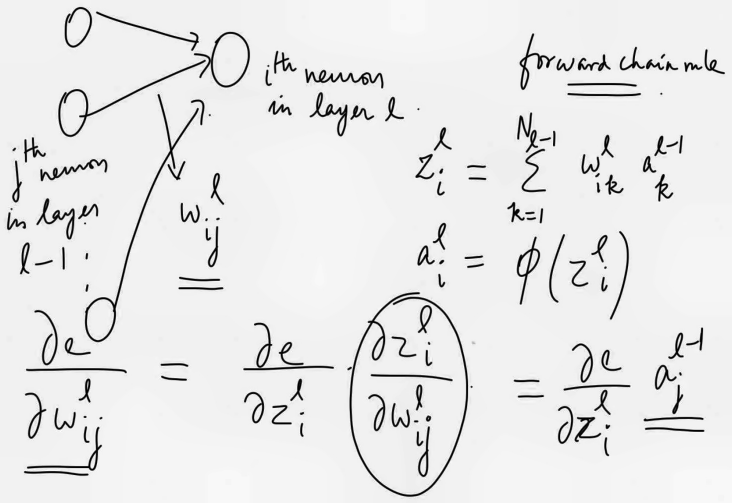


Figure 7: Two neurons in layer $l-1$ (j^{th} neuron, and one above it) connecting via weight w_{ij}^l to the ith neuron in layer l.

i^{th} neuron in layer l. j^{th} neuron in layer $l-1$.

forward chain rule

$$z_i^l = \sum_{k=1}^{N_{l-1}} w_{ik}^l a_k^{l-1}, \qquad a_i^l = \phi(z_i^l)$$

$$\frac{\partial e}{\partial w_{ij}^l} = \frac{\partial e}{\partial z_i^l}\left(\frac{\partial z_i^l}{\partial w_{ij}^l}\right) = \frac{\partial e}{\partial z_i^l} a_j^{l-1}$$

$$\frac{\partial e}{\partial z_i^l} = \frac{\partial e}{\partial a_i^l}\left(\frac{\partial a_i^l}{\partial z_i^l}\right) = \frac{\partial e}{\partial a_i^l} \cdot \phi'(z_i^l)$$

$$\frac{\partial e}{\partial a_i^l} = \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial z_m^{l+1}}\left(\frac{\partial z_m^{l+1}}{\partial a_i^l}\right) = \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial z_m^{l+1}} \cdot w_{mi}^{l+1}$$

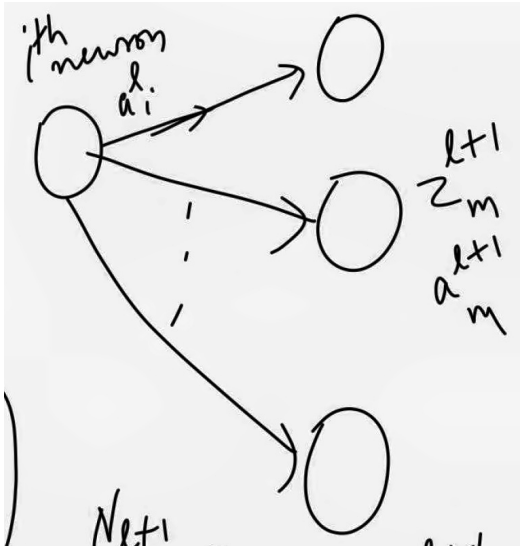


Figure 8: i^{th} neuron in layer l (activation a_i^l) connecting forward to multiple neurons in layer $l+1$ (activations z_m^{l+1}, a_m^{l+1}).

$$z_m^{l+1} = \sum_{k=1}^{N_l} w_{mk}^{l+1} \cdot a_k^l$$

$$\frac{\partial e}{\partial a_i^l} = \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial z_m^{l+1}} \cdot w_{mi}^{l+1}$$

$$= \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial a_m^{l+1}} \cdot \frac{\partial a_m^{l+1}}{\partial z_m^{l+1}} \cdot w_{mi}^{l+1} = \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial a_m^{l+1}} \phi'(z_m^{l+1}) \cdot w_{mi}^{l+1}$$

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